

ECE 647: Lecture 3

1 Affine Hull

The dictionary meaning of the word **hull** is: the husk, shell, or outer covering of a seed or fruit ¹.

Definition 1. *The affine hull of a set C is the smallest affine set that contains C . Denote it by $\text{aff}(C)$.*

Q) How to find the affine hull of C ?

A) Take any points in C . Their linear combination must belong to the affine hull.

Claim:

$$\text{aff}(C) = \{y | y = \theta_1 x_1 + \dots + \theta_k x_k, x_1, \dots, x_k \in C \text{ \& } \theta_1 + \dots + \theta_k = 1\}$$

Let us skip the proof as a similar proof will be discussed for convex hulls.

2 Subspaces and Affine sets

Closely related to affine sets is the concept of subspaces.

- A subspace is an affine set that passes through the origin
- Any affine set is a subspace plus an offset

Definition 2. *A set V is a subspace if for any $x_1, x_2, \dots, x_k \in V$, then*

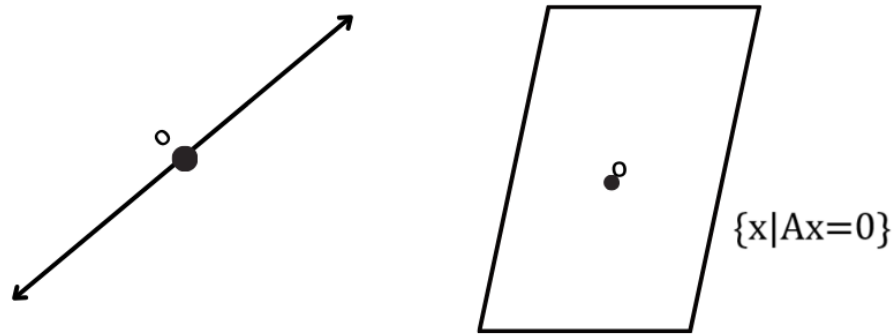
$$\theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k \in V$$

for any $\theta_1 \dots \theta_k$.

¹Pasted from <http://dictionary.reference.com/browse/hull>

Difference from affine sets

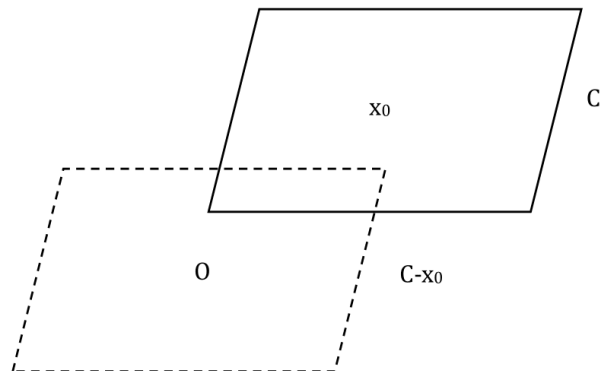
- We do not need $\theta_1 + \dots + \theta_k = 1$
- The origin must belong to a subspace



Claim: For any affine set C , pick a point $x_0 \in C$. Let,

$$C - x_0 = \{x - x_0 | x \in C\}$$

Then $C - x_0$ is a subspace.



Proof:

Assume that $V_1, V_2, \dots, V_k \in C - x_0$. Then,

$$V_1 = x_1 - x_0, V_2 = x_2 - x_0, \dots, V_k = x_k - x_0$$

for some $x_1, \dots, x_k \in C$.

Hence,

$$\begin{aligned} \theta_1 V_1 + \dots + \theta_k V_k &= (\theta_1 x_1 + \dots + \theta_k x_k) - (\theta_1 + \dots + \theta_k) x_0 \\ &= [(\theta_1 x_1 + \dots + \theta_k x_k) + x_0(1 - (\theta_1 + \dots + \theta_k))] - x_0 \\ &\in C - x_0 \end{aligned}$$

Summary

- Any affine set is a subspace plus an offset
- The subspace associated with the affine set C does not depend on the choice of x_0
- Often a subspace is the set of x such that $Ax = 0$, then the affine set is $Ax = b$.

3 Convex sets

Definition 3. A set C is convex if for any $x_1, \dots, x_k \in C$ we must have,

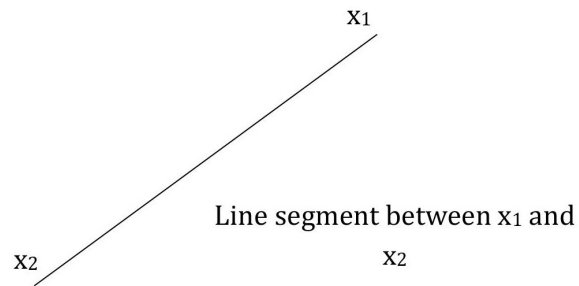
$$\theta_1 x_1 + \dots + \theta_k x_k \in C$$

for all $\theta_1 + \theta_2 + \dots + \theta_k = 1$ and $0 \leq \theta_1, \dots, \theta_k \leq 1$.

Also these combinations are called convex combinations.

Assume that $x_1, x_2 \in C$, for C to be a convex set we must have,

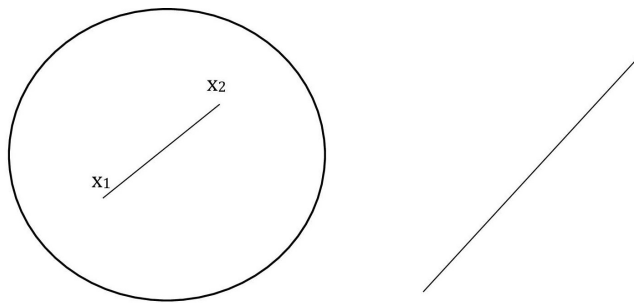
$$\begin{aligned} \theta_1 x_1 + (1 - \theta_1) x_2 &\in C \\ \Leftrightarrow x_2 + \theta_1(x_1 - x_2) &\in C, 0 \leq \theta_1 \leq 1 \end{aligned}$$



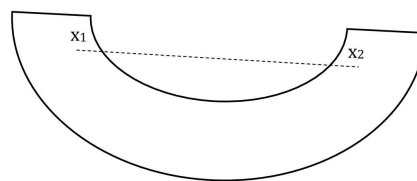
Roughly Speaking: A set is convex if you can go from every point to another point via a straight line without leaving the set.

Difference from affine sets: We only need the line segment that belongs to C . We don't need the part outside x_1 or x_2 .

Examples: A circle and a line segment are examples of convex sets.



This is not a convex set:



(A convex set does not have holes)

Q: Are affine sets convex?

Q: Can a convex set be an open set or a closed set?

4 Convex hull

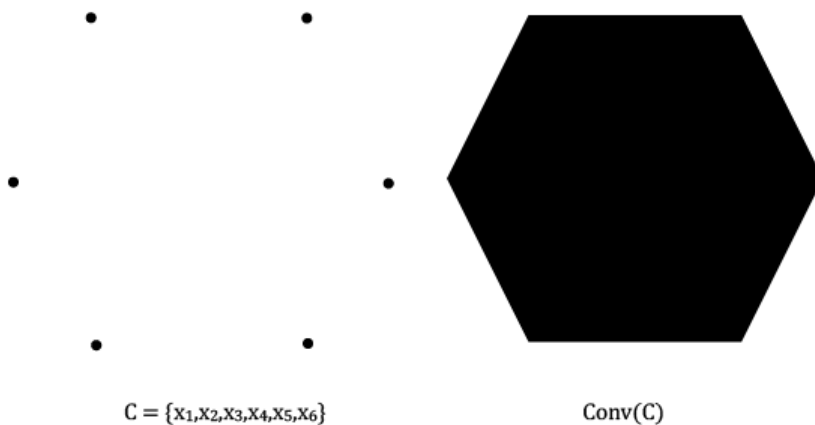
Definition 4. *The convex hull of a set C is the smallest convex set containing C .*

Take any two points $x_1, x_2 \in C$, then the line segment between x_1 and x_2 must belong to the convex hull.

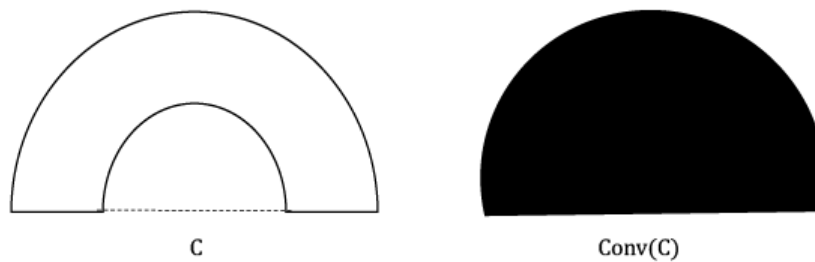
So we can construct convex hull in this way.

Examples:

1)



2)



Think of the region enclosed by a rubber band.

Claim: $\text{Conv}(C) = \{y | y = \theta_1 x_1 + \dots + \theta_k x_k, \text{ for some } x_1, \dots, x_k \in C, 0 \leq \theta_1, \theta_2, \dots, \theta_k \leq 1, \theta_1 + \dots + \theta_k = 1\}$

Denote the RHS as the set A .

Q: Why is it the smallest convex set containing C ?

A:

1. A is convex
2. For any convex set B containing C , B must contain A .

To show (1): We need to show that, if $V_1, \dots, V_j$ are all convex combinations of elements of C , then any convex combinations of $V_1, \dots, V_j$ is also a convex combination of elements of C .

To see this, assume

$$\begin{aligned} V_1 &= \theta_1^1 x_1 + \dots + \theta_k^1 x_k \\ V_2 &= \theta_1^2 x_1 + \dots + \theta_k^2 x_k \\ &\vdots \\ V_j &= \theta_1^j x_1 + \dots + \theta_k^j x_k \end{aligned}$$

For any $0 \leq \gamma_1, \gamma_2, \dots, \gamma_j \leq 1$ & $\gamma_1 + \dots + \gamma_j = 1$, we have,

$$\begin{aligned} \gamma_1 V_1 + \dots + \gamma_j V_j &= x_1(\gamma_1 \theta_1^1 + \gamma_2 \theta_1^2 + \dots + \gamma_j \theta_1^j) \\ &\quad + x_2(\gamma_1 \theta_2^1 + \gamma_2 \theta_2^2 + \dots + \gamma_j \theta_2^j) \\ &\quad + \vdots \\ &\quad + x_k(\gamma_1 \theta_k^1 + \gamma_2 \theta_k^2 + \dots + \gamma_j \theta_k^j) \end{aligned}$$

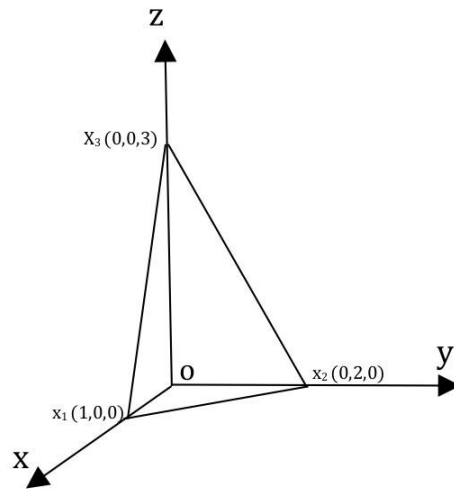
The coefficients are between 0 and 1 and their sum is equal to $\gamma_1 + \gamma_2 + \dots + \gamma_j = 1$.

To show (2): Trivially from the definition of Convex hull.

5 Algebraic forms

The beauty of the theory of convexity is that every notion has both an algebraic definition and a geometric interpretation. It is important to be able to go back and forth.

Q: What is the convex hull of the set containing $(1,0,0)$, $(0,2,0)$ and $(0,0,3)$ (think of this as the transmission rate to each user individually)?



A: It must be a triangle. The plane (subspace) that this triangle belongs to is:

$$\frac{x}{1} + \frac{y}{2} + \frac{z}{3} = 1$$

need constraints $x, y, z \geq 0$.

Alternatively, the convex hull is the set of y's such that,

$$\begin{aligned} y &= \theta_1(1, 0, 0) + \theta_2(0, 2, 0) + \theta_3(0, 0, 3) \\ 0 &\leq \theta_1, \theta_2, \theta_3 \leq 1, \theta_1 + \theta_2 + \theta_3 = 1 \end{aligned}$$

Can show that the two descriptions are equivalent.

$$y = (\theta_1, 2\theta_2, 3\theta_3)$$